

Module/Pathway Review: Apolipoprotein N-acyltransferase (UniPathway UPA00666) in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module/bucket:** unipathway:UPA00666 — "Protein modification; lipoprotein biosynthesis (N-acyl transfer)" **Primary gene:** lnt / PP_4790 / UniProtKB Q88DN4 **Iteration:** 1 of 3

1. Executive summary

- UPA00666 is a **single-step module** describing the final, phospholipid-dependent **N-acylation** of the N-terminal cysteine of a diacylglycerol-modified apolipoprotein, producing the mature triacylated lipoprotein. It is the third and last enzymatic step of the Lgt→LspA→Lnt lipoprotein maturation pathway.
- In *P. putida* KT2440 the module is **COVERED** by a single, high-confidence gene: **lnt / PP_4790 (Q88DN4)**, annotated as Apolipoprotein N-acyltransferase, EC 2.3.1.269, inner-membrane, CN-hydrolase family / apolipoprotein N-acyltransferase subfamily.
- There is **no paralog ambiguity**: PP_4790 is the *only* PSEPK protein bearing the Lnt-specific InterPro signature (IPR004563) and EC 2.3.1.269. Other CN-hydrolase (Pfam PF00795) members in the genome are metabolic nitrilases/amidases and are **not** over-annotated as N-acyltransferases.
- The surrounding machinery is intact — upstream **lgt/PP_5142** and **lspA/PP_0604**, and the downstream **Lol** sorting system (lolA/PP_4003, lolB/PP_0724, lolCDE/PP_2154–2156) — supporting functional coherence.
- Confidence is further upgraded by **catalytic-residue verification**: PP_4790 shares **44% identity** with *E. coli* Lnt over the full length and conserves the **entire catalytic apparatus** — triad Glu267→Glu269, Lys335→Lys329, Cys387→Cys381 (nucleophile), plus accessory residues Glu343→Glu337, Trp237→Trp240, Tyr388→Tyr382, Glu389→Glu383 (this review's alignment).

- Principal caveat: the PSEPK Lnt annotation is **homology-inferred** (UniProt protein existence level 3); there is no direct enzymological or genetic characterization *in P. putida KT2440* itself. Confidence is nonetheless high because the family assignment is specific, the catalytic machinery is fully conserved, and the pathway context is complete.

Recommendation: mark the single module step **covered**; no gap, no revision needed. Promote PP_4790 to full `fetch-gene` review only for provenance/essentiality confirmation, not because of annotation doubt.

2. Target-organism pathway definition

What the module includes (and only this): transfer of one acyl chain from a *sn*-1 glycerophospholipid (e.g., phosphatidylethanolamine) to the free α -amino group of the N-terminal S-diacylglyceryl-cysteine of an apolipoprotein, yielding an amide-linked (N-acyl) fatty acid and the mature triacylated lipoprotein. UniProt catalytic definition for Q88DN4:

N-terminal S-1,2-diacyl-*sn*-glyceryl-L-cysteiny-[lipoprotein] + a glycerophospholipid $\rightarrow$ N-acyl-S-1,2-diacyl-*sn*-glyceryl-L-cysteiny-[lipoprotein] + a 2-acyl-*sn*-glycero-3-phospholipid + H⁺ (EC 2.3.1.269)

Boundaries — keep separate: - **Lgt step** (PP_5142; EC 2.4.99.65-type diacylglyceryl transfer) and **LspA step** (PP_0604; signal peptidase II, EC 3.4.23.36). These are the two *upstream* steps of lipoprotein maturation and belong to their own UniPathway steps, not UPA00666. - **Lol sorting / outer-membrane targeting** (LolABCDE) — a *downstream transport* process, not part of the N-acyl-transfer biochemistry. - **General "lipid/phospholipid metabolism" overview maps** — Lnt consumes a phospholipid as acyl donor but is a protein-modification enzyme, not a lipid-biosynthesis enzyme; do not merge into fatty-acid/phospholipid metabolism buckets. - **Lnt-like N-acyltransferases acting on non-lipoprotein substrates** (e.g., AatD of Enterobacterales, which acylates the anti-aggregation protein Aap;).

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These are structural relatives, not part of canonical lipoprotein maturation, and are a known source of mis-transfer of the "Lnt" label — but no such gene is present in PSEPK.

Alternate names / database mappings: Lnt; apolipoprotein N-acyltransferase; "ALP N-acyltransferase"; EC 2.3.1.269; KEGG ppu:PP_4790; eggNOG/COG0815; Pfam PF00795 (CN_hydrolase) + PF20154; InterPro IPR004563 (subfamily-specific), IPR003010, IPR036526, IPR045378; PROSITE PS50263; GO:0016410 (N-acyltransferase activity), GO:0042158 (lipoprotein biosynthetic process), GO:0005886 (plasma membrane).

3. Expected step model

UPA00666 as scoped is a **one-step module**:

Step	Reaction	Expected enzyme	PSEPK gene	Status
S1	Apolipoprotein → mature triacylated lipoprotein (N-acyl transfer)	Lnt (EC 2.3.1.269)	lnt / PP_4790 (Q88DN4)	covered

Contextual (out-of-module but confirming) steps of the wider maturation pathway, all present in PSEPK: - Step -2 (Lgt): **PP_5142 (Q88CN8)**, phosphatidylglycerol-prolipoprotein diacylglycerol transferase. - Step -1 (LspA): **PP_0604 (Q88Q91)**, lipoprotein signal peptidase II. - Downstream (Lol): PP_4003, PP_0724, PP_2154-2156.

No step is "missing but likely present under another name," because the module contains exactly one step and it is directly encoded. No step is "not expected" here.

4. Candidate genes and evidence

lnt / PP_4790 / Q88DN4 — HIGH confidence

- **Role:** final N-acylation step of lipoprotein maturation (EC 2.3.1.269); inner-membrane integral protein, 505 aa.
- **Evidence type:** sequence homology / family assignment (UniProt existence level 3 = "inferred from homology"). Family: **CN hydrolase → apolipoprotein N-acyltransferase subfamily**; Lnt-specific **InterPro IPR004563**, **PROSITE PS50263**, **COG0815**. UniPathway

UPA00666 and pathway comment "Protein modification; lipoprotein biosynthesis (N-acyl transfer)" both attached.

- **Mechanistic anchor (from orthologs, transfer to PSEPK strong):** *E. coli* Lnt proceeds via a covalent **thioester acyl-enzyme intermediate** on an active-site cysteine (Cys387), with a Glu–Lys–Cys catalytic triad (Glu267 required for acyl-enzyme formation) and a ping-pong mechanism with phospholipid head-group/acyl-chain preference (PMIDs 20000742, 21676878). Structural basis of the reaction has been solved for a bacterial Lnt

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- **Catalytic-residue verification (this review, direct sequence evidence for PP_4790):** global BLOSUM62 alignment of Q88DN4 vs *E. coli* Lnt (P23930) gives **44.0% identity over 484 aligned positions**, and every functionally essential *E. coli* residue is conserved in PP_4790 — catalytic triad **Glu267→Glu269, Lys335→Lys329, Cys387→Cys381**; accessory residues **Glu343→Glu337, Trp237→Trp240, Tyr388→Tyr382, Glu389→Glu383** (roles per

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This makes the mechanistic transfer to *P. putida* **strong at residue resolution**, not merely fold-level.

- **Curation caveats:**
- Annotation is homology-based; there is **no direct *P. putida* KT2440 biochemistry or knockout**. Returned essentiality/Tn-seq screens (e.g.,

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did not explicitly report PP_4790 status in the abstracts examined.

- The broad Pfam PF00795 fold is shared with unrelated nitrilases/amidases; rely on the **subfamily-level IPR004563 / EC 2.3.1.269**, not PF00795 alone, when asserting function.
- EC 2.3.1.269 is highly specific (dedicated to this reaction), so the EC→function mapping here is safe (low over-broad-EC risk).

5. Gaps, ambiguities, and likely over-annotations

- **No gaps.** The single expected step is encoded.
- **No paralog ambiguity.** Exactly one IPR004563 / EC 2.3.1.269 protein exists in PSEPK (PP_4790). The other eight PF00795 CN-hydrolase members — e.g., YafV omega-amidase (PP_0859, EC 3.5.1.3), davA (PP_0382), ramA (PP_3846), plus uncharacterized CN-hydrolase-domain proteins (PP_3019, PP_0896, PP_4463, PP_2005, PP_0939) — are metabolic hydrolases and are **not** mis-labeled as N-acyltransferases. No pruning needed.
- **Over-annotation risk elsewhere (not in PSEPK):** the AatD-type Lnt-like N-acyltransferases of Enterobacterales acylate non-lipoprotein substrates

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Curators should watch for this pattern in *other* genomes but it does not affect *P. putida* KT2440.

- **Essentiality nuance:** in most γ -proteobacteria with a single Lol pathway (*E. coli*, *Salmonella*), *lnt* is **essential** (PMIDs 15513925, 21742870). Exceptions (*Francisella*, *Neisseria*) possess modified LolCDE able to sort diacylated lipoproteins and tolerate *lnt* loss

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P. putida KT2440 has a canonical LolCDE (PP_2154–2156), so *lnt* is **predicted essential**; this is an inference, not a direct KT2440 measurement.

6. Module and GO-curation recommendations

- **Step S1 (N-acyl transfer):** mark **covered** by PP_4790.
- **Module status: satisfiable / complete.** No `gap`, `candidate_uncertain`, `not_expected_in_target_taxon`, or `module_needs_revision` flags warranted.
- **Module boundaries:** the generic single-step scope is **correct** for this organism. Keep Lgt (PP_5142) and LspA (PP_0604) in their own upstream module steps; keep Lol sorting separate.
- **GO terms:** existing assignments (GO:0016410 N-acyltransferase activity; GO:0042158 lipoprotein biosynthetic process; GO:0005886 plasma membrane) are appropriate. A more specific MF term "apolipoprotein N-acyltransferase activity" would better match EC 2.3.1.269 if available/desired, but no new GO term request is strictly required.

- **No new module document needed.**

7. Genes to promote to full review

- **PP_4790 (Int, Q88DN4):** the functional assignment is now well-supported (specific IPR004563 signature **and** full catalytic-triad conservation verified in this review), so promotion is **optional and low-priority** — useful only for provenance/essentiality confirmation (check KT2440 Tn-seq/CRISPRi essentiality datasets and add experimental-existence evidence if available). This is the only gene in the module.
- No other PSEPK gene needs promotion for this module.

8. Key references

- Robichon C, et al. (2005) Depletion of Lnt mislocalizes OM lipoproteins in *E. coli*; Lnt essential, 6 TM segments.

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- Buddelmeijer N, Young R (2010) Lnt acts via an extracytoplasmic thioester acyl-enzyme intermediate; Glu–Cys catalytic residues (Cys387, Glu267).

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- Hillmann F, et al. (2011) Kinetics and phospholipid specificity of Lnt; ping-pong mechanism.

P 21676878

- Narita S, Tokuda H (2011) LolCDE overexpression permits *lnt* deletion — links Lnt to Lol sorting.

P 21742870

- LoVullo ED, et al. (2015) *lnt* non-essential in *Francisella/Neisseria* with modified LolCDE — two classes of Gram-negatives.

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- Noland CL, et al. (2017) Structural insights into lipoprotein N-acylation by Lnt.

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- Nozeret K, et al. (2019) Sensitive fluorescence assay for Lnt activity.

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- Belmont-Monroy L, et al. (2020) AatD, an Lnt-like N-acyltransferase acting on non-lipoprotein Aap — over-annotation caution.

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- Brülle JK, et al. (2013) Lnt in slow-growing mycobacteria; distribution of Lnt across GC-rich taxa.

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- UniProtKB **Q88DN4** (Lnt, PP_4790); KEGG **ppu:PP_4790**; UniPathway **UPA00666**; InterPro **IPR004563**.

9. Evidence provenance and open questions

Direct for PSEPK (database/sequence evidence): gene identity, family/InterPro/EC assignment, genomic presence of the full Lgt/LspA/Lnt + Lol machinery, single-copy status (no paralog), and the 44%-identity residue-level conservation of the catalytic triad in PP_4790 (this review's alignment vs *E. coli* P23930).

Transferred from other bacteria (homology-based; strength noted): the two-step ping-pong mechanism, thioester acyl-enzyme intermediate, phospholipid specificity, and essentiality/dispensability logic — all from *E. coli* and other Gram-negatives (strong transfer for mechanism given residue conservation; moderate for essentiality given differing Lol architectures across taxa).

Open questions / experiments that would close remaining gaps: 1. **Direct essentiality in KT2440** — targeted literature searches in this review did not surface a KT2440-specific *lnt* essentiality call. A CRISPRi knockdown or attempted markerless deletion (with/without LolCDE overexpression) would confirm the predicted-essential status. 2. **Experimental protein existence** — PP_4790 is currently existence-level 3 (homology); proteomic or activity evidence (e.g., the fluorescence Lnt assay,

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would raise it to experimental. 3. **Lipoprotein acylation state** — MS of a model KT2440 outer-membrane lipoprotein would confirm triacylation (N-acyl transfer actually occurring in vivo).

No wet-lab characterization of PP_4790 in P. putida KT2440 was found in this review; the functional call rests on specific family signatures plus full catalytic-residue conservation, which together constitute strong (but indirect) evidence.

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